

IR-RAG @SIGIR25: The Second Edition of the Workshop on Information Retrieval's Role in RAG Systems

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Abstract

In recent years, Retrieval-Augmented Generation (RAG) systems have become a cornerstone of artificial intelligence, attracting considerable attention in a variety of fields. By integrating the strengths of information retrieval and generative models, these systems have shown immense potential to push the boundaries of machine learning applications. Nevertheless, RAG systems still face significant challenges and offer ample room for advancement and innovation.

This workshop aims to highlight the central role of information retrieval within RAG frameworks, which we believe has often been overshadowed by the emphasis on the generative components. While the generative models are integral to these systems, the quality and effectiveness of the retrieval mechanism is equally critical, as it has a direct impact on the overall system performance and outcomes. We invite papers that rethink and prioritise the fundamental aspects of RAG systems, particularly in strengthening the information retrieval component. Through this workshop, we aim to gain deeper insights into how improved retrieval methods can enhance the performance and reliability of RAG systems. The event will bring together leading experts, researchers and practitioners to provide a collaborative platform for exchanging ideas, sharing results and fostering innovation. Our aim is to stimulate research and discussion that reaffirms the essential role of information retrieval in shaping the next generation of generative systems.

CCS Concepts

• **Information systems** → *Novelty in information retrieval.*

Keywords

Retrieval Augmented Generation, Generative Models, Neural Databases

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1 Motivation

In the rapidly evolving fields of artificial intelligence and machine learning, RAG systems have emerged as a transformative innovation [9]. Designed to improve language models [8, 15] through the integration of external knowledge, RAG systems have proven invaluable in a wide range of applications, including natural language processing and recommendation systems. By leveraging knowledge repositories, these systems generate coherent, context-aware outputs that push the boundaries of AI's ability to emulate human reasoning. Key attributes of RAG systems that align with human cognition include integration of external knowledge, contextual understanding and adaptive learning.

Given these strengths, a focused workshop on the retrieval processes within AI systems is timely and essential. Such a workshop would explore the mechanisms that enable effective retrieval and integration of diverse knowledge sources, with the aim of advancing AI systems towards greater adaptability and complexity that more closely mirrors human cognitive abilities.

Despite their impressive capabilities, RAG systems face significant challenges. Chief among these is their reliance on the quality and relevance of retrieved documents. Irrelevant or biased sources can produce misleading results [7], although in some cases randomly retrieved documents have paradoxically improved performance [5]. In addition, these systems often struggle to maintain consistency and coherence in their responses, occasionally producing contradictory or unclear results [6].

While some advances show promise in dealing more effectively with large contexts [14], fundamental problems remain, such as the lost in the middle effect [10], where the performance of language models deteriorates when critical information appears in the middle of a context. In addition, reliance on external databases introduces the challenge of obsolescence, as outdated databases can lead to factually incorrect responses over time [13]. The retrieval process also adds computational complexity, increasing resource consumption and response latency. Privacy concerns further complicate matters, with the risk of exposing sensitive data or violating privacy policies when accessing external data [4]. Broadly speaking, the challenges of RAG systems can be divided into two areas: limitations in the retrieval process and issues within the generative

component. However, much of the discourse surrounding RAG systems has focused disproportionately on the generative side, often overlooking the equally crucial role of information retrieval. Our workshop aims to address this imbalance by focusing on the retrieval mechanisms that underpin RAG systems. While advances in large language models continue to attract considerable attention for their impressive capabilities, the retrieval processes that feed these models remain underexplored. By prioritising research into these fundamental mechanisms, we hope to drive advances that will improve the overall efficiency and performance of RAG systems, unlocking new potential in their application and development.

2 Theme and purpose of the workshop

The primary goal of this workshop is to bring much-needed attention to the retrieval mechanism within RAG systems, while exploring the central question: “How should information retrieval research evolve in the era of RAG systems?”. By bringing together experts, practitioners and enthusiasts, the workshop aims to foster dialogue and innovation around the challenges and opportunities associated with the retrieval component of RAG systems. The goal is to cultivate a robust community dedicated to advancing this critical aspect of RAG, ensuring that it receives equal focus alongside the generative elements. Through this platform, we aim to share knowledge, inspire new research directions, and foster collaborative efforts to better understand and improve retrieval mechanisms. Ultimately, we envision building a strong foundation for a thriving community committed to shaping the future of information retrieval for RAG systems. We invite papers that explore a variety of topics, including but not limited to

Use Of The Retrieved Context By The LLM: Studies [5, 9, 12] have shown that RAG systems are sensitive to type and order of retrieved contexts. These findings open avenues for deeper exploration of how retrieved information influences generative output.

(Query) Representation Learning: Improving queries’ representation can significantly enhance a retriever’s ability to find relevant documents. This includes using advanced NLP techniques to better capture the contextual subtleties and intent behind queries.

Incorporating Contextual Information: Incorporating broader contextual elements (e.g. conversation history, user preferences) into the search process can lead to more accurate results.

Updating the Document Database: Research into efficient update mechanisms and real-time synchronisation, which ensures that document repositories are up to date, is encouraged.

Reducing Computational Load: Optimising retrieval methods for speed and efficiency, especially on large datasets, can make these systems more practical for real-time applications.

Bias Mitigation: Investigating ways to identify and reduce bias in retrieval processes is crucial to ensure the fairness and reliability of the content generated.

Cross-Lingual Retrieval Capabilities: Improving retrieval capabilities in multilingual environments is essential for systems serving diverse audiences.

Multimodality: While most research has focused on text-based RAG systems, multimodal capabilities are increasingly important for a wide range of applications. Research into methods for incorporating and retrieving data from multiple modalities is encouraged.

3 Format and Planned interaction

This workshop will take place over a full day and will include a variety of activities designed to encourage engagement and collaboration among participants. The format will include **(a)** Renowned speakers from leading companies and institutions will deliver keynote speeches; **(b)** Selected high quality papers will be presented in oral sessions; **(c)** Papers with less emphasis will be presented as posters to encourage discussion and feedback; **(d)** Focused group discussions on topics inspired by contributed papers and broader workshop themes, will provide an opportunity for knowledge exchange.

Event	Date
Submission deadline	April 23, 2025
Notification	May 21, 2025
IR-RAG Workshop	July 17, 2025

Table 1: Important dates

Duration	Activity
15min	Opening
1h	Keynote speaker (1)
45min	Paper presentations (1)
30min	Coffee break
45min	Paper presentations (2)
1h30min	Lunch break
1h	Keynote Speaker (2)
45min	Poster Session
30min	Refreshment break
1h	Breakout session & discussion among participants
30min	Round up and concluding remarks

Table 2: Schedule

Tables 1 and 2 list the workshop schedule and key dates.

4 Distinction from main conference

The proposed workshop is designed to complement the main conference programme by focusing on areas that are often underrepresented or overlooked in standard sessions. While the conference typically highlights advances in core information retrieval technologies, our workshop narrows its scope to the interplay between retrieval mechanisms and generation processes within RAG systems. This theme includes emerging, niche, and interdisciplinary topics that offer fresh perspectives, such as the optimisation of retrieval pipelines, the integration of multimodal data, and cross-lingual retrieval for generative models. In addition, the workshop aims to address practical challenges such as computational efficiency, database updates, and bias mitigation, which are critical to real-world applications but rarely the focus of mainstream conference discussions.

5 Organizers attendance

With the exception of Ziheng Chen, all organizers intend to participate in the SIGIR conference in person and will be available

onsite if the workshop proposal is approved. The workshop does not require any special arrangements or accommodations.

6 Organizers

Negar Arabzadeh is a PhD candidate at the University of Waterloo, supervised by Dr. Charles Clarke. She has been conducting research in Information Retrieval and Natural Language Processing for over six years as a graduate student and research assistant at Toronto Metropolitan University and the University of Waterloo. Her research interests lie in Ad-hoc Retrieval, specifically the evaluation of IR systems. Negar has presented tutorials on fairness and evaluation in information retrieval at SIGIR 2022, WSDM 2022, and ECIR 2023 and 2024. She has also completed research-oriented internships at Microsoft Research, Spotify Research, and Google Brain, working on various projects related to IR and NLP. Additionally, she was one of the lead organizers of the NeurIPS IGLU competition on Interactive Grounded Language Understanding in a Collaborative Environment. Negar has been recognized with multiple awards, including best paper awards at ECIR 2022 and ECIR 2024. She has also been honored as a distinguished referee at CIKM 2021 and 2023 and received travel awards for several conferences.

Ziheng Chen is currently a Senior Data Scientist at Walmart Global Tech, USA, where he is part of the Personalization Research group. His work focuses on developing trustworthy and explainable recommender systems. Ziheng has published research in Explainable Machine Learning, Reinforcement Learning, and Recommender Systems at leading conferences such as SIGIR, CIKM, and IJCNN, as well as in journals including IEEE TAI, Machine Intelligence Research, and Chem. He has also served as a Program Committee member for conferences such as NeurIPS, CIKM, SIGIR and ECML, and as a reviewer for journals including IEEE TAI, ACM TIST, and ACM TOIS. He earned his Ph.D. in Applied Mathematics and Statistics (AMS) from Stony Brook University in 2022, with a dissertation on Explainable AI. His thesis was recognized with the WOO-Jong Kim Award for the best dissertation in AMS.

Fabio Petroni is co-founder at Samaya AI, a company building an AI-powered knowledge-discovery platform. Before that, he was Researcher at the FAIR team of Meta AI and the R&D department at Thomson Reuters, where he focused on Knowledge-Intensive Natural Language Processing. He was the Sponsor Chair for EuroSys 2023, Local Chair for AKBC 2022, and co-organized the Workshop on Representation Learning for NLP hosted at ACL 2020. He is one of the authors of the original paper on RAG [9], of the Lost in the middle paper [10] and Autoregressive Search Engines [3].

Federico Siciliano is a PostDoc in Data Science at Sapienza University of Rome. He is part of the RSTLess research group at the Sapienza University of Rome, which focuses on Robust, Safe, and Transparent Deep Learning. Siciliano has publications in the domains of Information Retrieval and Explainable Artificial Intelligence, featuring conferences such as ECIR, RecSys, and IJCNN. He has also been a Program Committee member for SIGIR, ECIR, RecSys, AAAI and NeurIPS for multiple years. Siciliano is a Ph.D. Candidate in Data Science from the University of Sapienza, with a thesis on "Architectural Components of Trustworthy Artificial Intelligence". He is one of the authors of RRAML [1].

Fabrizio Silvestri is a Full Professor at Sapienza University of Rome. He has authored more than 150 papers in international journals and conference proceedings and holds nine industrial patents. Silvestri has been recognized with a "test-of-time" award at the ECIR 2018 conference for an article published in 2007. He also received three best paper awards and other international recognitions. Silvestri spent eight years in industrial research laboratories, including Yahoo! and Facebook. At Facebook AI, he directed research groups to develop artificial intelligence techniques to combat malicious actors who use the Facebook platform for malicious purposes, such as hate speech, misinformation, and terrorism. Silvestri has experience in organizing numerous workshops and conferences, and he will be one of the General Chairs of ECIR 2025 in Lucca and one of the Program Committee Chairs of CIKM 2026 in Rome. Silvestri holds a Ph.D. in computer science from the University of Pisa, with a thesis on "High-Performance Issues in Web Search Engines: Algorithms and Techniques". He is one of the authors of Multimodal Neural Databases [16] and RRAML [1].

Giovanni Trappolini is an Assistant Professor at the Department of Computer, Control and Management Engineering at Sapienza University of Rome. He received his Ph.D. in Machine Learning in 2022 under the supervision of Professor Emanuele Rodolà and was awarded the title of Sapienza's honor student. His research interests lie at the intersection of geometric and multimodal deep learning, with a particular focus on information retrieval. He has a solid track record of publications in the most important machine learning conferences and has the privilege of partnering with numerous world-renowned institutions, including Stanford, Technion, Meta, Amazon, and UniPi, among others. He is the creator of Fauno [2], one of the largest (and best-performing) Italian large language model to date, and one of the authors of Multimodal Neural Databases [16] and The Power of Noise [5]. Last year, he also organized the FLIRT [11] workshop at SIGIR.

7 Potential PC members

- Aditya Sanghi, University Of Toronto | Autodesk
- Aleksandra Piktus, Sapienza University of Rome | Cohere
- Alon Halevy, Meta
- Bora Edizel, Warner Bros. Discovery
- Charles L. A. Clarke, University of Waterloo
- Franco Maria Nardini, ISTI-CNR
- Iftah Gamzu, Amazon
- James Thorne, KAIST
- Konstantina Palla, Spotify
- Nicola Tonello, University of Pisa
- Prabhat Agarwal, Pinterest
- Raffaele Perego, ISTI-CNR
- Ricardo Baeza-Yates, Pompeu Fabra University
- Shoval Lagziel, Amazon
- Wang-Chiew Tan, Meta

8 Selection process

The workshop invites submissions of papers ranging from two to nine pages via an open call for contributions. We welcome diverse submissions, including reports detailing original research, preliminary research findings, proposals for innovative work, and position

papers. All submitted papers will undergo a rigorous peer review process led by the Program Committee.

9 Target audience

This workshop aims to attract a broad and diverse audience, including researchers, industry professionals, graduate students, and enthusiasts interested in RAG systems as well as the broader fields of IR and Generative Machine Learning (GML). It welcomes participants from academia, industry, and beyond who are keen to explore the intersection of these domains. To ensure extensive outreach, the workshop will be promoted through a range of channels. We will utilize established academic networks, including collaborations with research institutions and dissemination via key academic mailing lists. Direct email invitations will be sent to researchers and practitioners within the field, encouraging submissions and participation. In addition, the workshop will be actively promoted on social media platforms such as Twitter, LinkedIn, and other relevant networks to reach a wider audience. A dedicated workshop website will act as a central repository of information, offering details about the event, submission guidelines, the schedule, keynote speakers, and other updates. This website will ensure that all interested participants can easily access comprehensive information about the workshop. By combining these targeted promotional efforts, the workshop seeks to engage a wide audience and foster a dynamic discussion around the integration of IR and GML.

10 Past Editions

The inaugural IR-RAG workshop at SIGIR 2024 demonstrated significant interest in this emerging research domain. The workshop achieved maximum participation among SIGIR 2024 workshops, with attendance metrics indicating 80-100 concurrent participants. Post-workshop evaluation derived from participant feedback confirmed the interest in the themes of the workshop. Furthermore, the workshop concluded with a collaborative breakout session, during which participants identified critical research directions and emphasized the imperative for sustained investigation in this domain. This consensus among domain experts regarding the significance of advancing IR techniques for RAG systems provides substantial justification for the workshop's continuation.

11 Related workshops

Generative IR Workshops, such as Gen-IR@SIGIR24 have primarily explored the application of generative methods within the realm of IR. To articulate the distinction, one might adapt President Kennedy's famous antimetabole: Generative IR asks what generation can do for IR; we ask what IR can do for generation. These workshops have often focused on techniques like Differentiable Search Index (DSI). In contrast, our workshop emphasizes not just generative techniques in IR but the enhancement of Information Retrieval methods to empower and extend the capabilities of Generative Machine Learning models. Retrieval-Enhanced Machine Learning (REML@SIGIR2023) workshop addressed broader themes in machine learning, whereas our proposed workshop hones in specifically on the critical role of retrieval in augmenting Generative Machine Learning models. Rather than focusing solely on general machine learning advancements, our workshop spotlights the

foundational importance of retrieval mechanisms within generative frameworks. These distinctions underscore the unique niche our workshop occupies. Despite overlaps with related workshops, there remains a clear gap in events dedicated explicitly to the retrieval aspect of RAG models. By addressing this gap, our workshop seeks to fill an essential need within the research community, focusing on the indispensable role of retrieval in refining the performance and utility of generative systems.

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